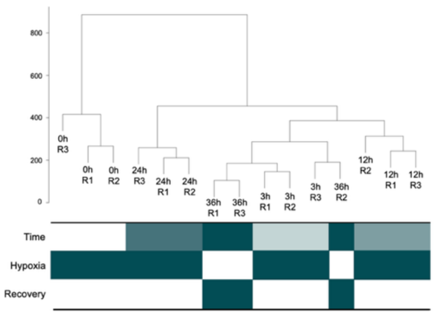
CANDIDATE CORRECTIONS TO MANUSCRIPT

- [Page 1]:
  - Across Eumetazoa, oxygen-sensing and hypoxia-responsive regulation **are** coordinated by the conserved hypoxia-inducible factor (HIF) pathway.
  - **Anthropogenic climatic changes are contributing to increases in the incidence of severe hypoxia in aquatic ecosystems**.
  - In this study, we show that the phylum Platyhelminthes has likely lost core components of the HIF pathway, yet still survives hypoxic stress and **mounts robust gene regulatory responses**.
  - **Nonetheless**, we find that planarians are tolerant to severe hypoxia for at least 24 hours.
  - An oxygen-dependent degradation (ODD) domain containing a proline residue within an 'AP' or 'SP' motif across phyla recognised by the HIF prolyl hydroxylase EGLN.
- [Page 3]:
  - Causing decreases in **metabolic rate**, locomotion, survival, growth or reproduction (Han & Hagiwara, 2023; Lee *at al.*, 2023; Vaquer-Sunyer *et al.*, 2008).
  - This led us to examine the gene regulatory response to hypoxia to begin to understand how **low oxygen stress** is managed in the absence of the canonical HIF pathway.
- [Page 4]:
  - Figure 1: Phylogeny of metazoan species from which **HIF pathway gene** anchor sequences etc.
  - Protein sequences from 25 species across 13 animal phyla were used to construct **p-HMMs for HIF-α, EGLN and VHL ortholog mining** across the proteomes of 9 platyhelminth species.
  - **Priapulida clades with Ecdysozoa**, not Lophotrochozoa.
- [Page 5]:
  - Figure 2: ***Taenia crassiceps*** was also sampled at the proteome level in Cestoda.
- [Page 14]:
  - **For HIF-α ortholog mining**, alignment using MAFFT iterative local-pair refinement was selected to maximise positional homology etc.
- [Page 16]:
  - Subsequent maximum-likelihood phylogenetic inference placed the *Clonorchis* sequence **in a phylogenetic tree of high-confidence metazoan VHL sequences** as an isolated branch outside other metazoan VHL clades etc.
- [Page 21]:
  - Figure 11A: **Amended**



  - Figure 11C: Heatmaps **beside** the module assignments etc.
  - Figure 11D: Temporal trajectories of module expression showing mean FPKM **value** of genes etc.
- [Page 31]: These indirect sources of evidence are critical in a non-model metazoan with no transgenic technologies available, precluding the use of **CRE–reporter** constructs for directly testing the function and mapping the target genes of CREs.
- [Page 32]: Figure 18: Base-pair resolution view of hypoxia-specific Klf10/11 footprint in **h1SMcG0021003**.